

Scale Buys Interpolation, Structure Buys a Horizon: Certified Predictability for Equivariant World Models

How far a world model can promise — tightly, and only with structure.

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Abstract

Scale buys interpolation; structure buys a certified horizon. A world model’s average error says nothing about whether a *particular* prediction can be trusted, or for how long. For **equivariant** latent world models we give a computable, multi-step **certificate of the predictable horizon**: T -step rollout error is provably constant over each symmetry orbit (Theorem A), stratified channel-by-channel by the predictor’s Lyapunov spectrum, $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$, and **two-sided** — a matching lower bound makes approximate equivariance provably horizon-limited (Proposition 6). The certificate is *equivalent* to equivariance (Lemma 2): no non-equivariant model has it at any scale. Empirically the law lifts to learned models of real chaotic systems — on 40-D Lorenz-96 only a $\mathbb{Z}_N$ -equivariant network recovers the full spectrum ($R^2=0.98$; dense and recurrent baselines fail). A cone certificate (Theorem B) then makes the horizon **a-priori sound**, tight exactly on uniformly-hyperbolic dynamics. The certificate is *actionable* — **a priori, with zero calibration data**: under a fixed sensing budget a $c\times$ -inflated certificate provably needs $c\times$ the budget (Proposition 9), and it **audits public pretrained world models training-free** — calibrated where strongly expansive (ratio 0.94–1.02), optimistic where weakly expansive, correctly abstaining where contracting — a scope map that a deployed sensing-budget monitor then **replicates cell-by-cell, out-of-sample, at zero new estimation**. Across the official 1M–317M multitask scale ladder, calibration does **not** improve with parameters — scale buys interpolation, not a calibrated horizon. §6 states the 1–2-GPU scope.

1. Introduction

A world model is judged by its average prediction error. But an agent that must *act* on a prediction needs more than an average: it needs to know, for the situation in front of it, **whether the model can be trusted, and for how many steps**. Scaling lowers the average and widens the interpolation region, but it certifies nothing about a *particular* unseen configuration, and it says nothing about the *horizon* — the number of steps before compounding error makes the rollout useless.

Equivariance is known to give *some* per-situation guarantee. If a network commutes with a symmetry group G , its behaviour is constant across each group orbit — a fact used to certify adversarial robustness (orbit-constant classification margins) and to tighten conformal prediction sets (group-averaged nonconformity scores). But every such result is **single-shot**: it attaches a guarantee to one input, for one prediction. A world model predicts a *trajectory*, and the question that matters for planning — *how far ahead is the guarantee valid?* — has no answer in that literature.

This paper answers it. We show that an equivariant latent world model admits a **certificate of its predictable horizon**, stratified channel-by-channel by the predictor’s spectrum, and we prove the horizon is **tight**. The contributions are:

1. **A tight certified horizon (§3.2)**. For an equivariant model the T -step rollout error is orbit-constant (Theorem A); relaxing to approximate equivariance, channel j with Lyapunov exponent λ_j is certified

to horizon $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$ (Theorem B, the *upper* bound). Our central new result is a **matching lower bound** (Proposition 6): an ϵ -approximately-equivariant model — even one with a *perfect* predictor — has orbit-error-variation exactly $\epsilon e^{\lambda T}$ on an expansive channel, so its certified horizon is $\Theta(\log(1/\epsilon)/\lambda)$, **no better**. Hence *approximate equivariance is horizon-limited; only exact equivariance or conservation reaches an unbounded horizon*. A **scope theorem** (Proposition 7) then says *when* the locally-measured spectrum governs a *learned* model’s horizon — Oseledets gives the rate on spectrally non-degenerate dynamics, the law degenerates on near-neutral dynamics — and a **finite-horizon continuity bound** (Proposition 8) makes the learned-model lift rigorous, not merely empirical. The horizon staircase then **lifts to learned models of a class of genuinely chaotic systems** (E2: Lorenz, Hénon, Rössler — the learned model’s Lyapunov exponent matches the true one to 1–12%), while the near-neutral PushT interior is the predicted degenerate case.

2. **The certificate is exclusive to structure (§3.1).** The converse (Lemma 2): orbit-constant error against every equivariant target $\iff$ the model is equivariant. So **no non-equivariant model has the certificate at any size** — an architectural impossibility, proved, not asserted.
3. **The Noether hinge (§3.3).** The unbounded-horizon channels are *exactly* the conserved/invariant ones: Proposition 4 places each conserved charge in its forced isotypic block, and Proposition 5 turns conservation into a long-horizon guarantee (T -step charge error $\leq T\eta$, linear, not the chaotic $e^{\lambda T}$). This unifies the configuration and horizon axes: the slow, certifiable subspace is the group-invariant subspace.
4. **Single-shot certified equivariance is our $T=1$ slice (§5).** We position the robustness/conformal guarantees as the one-step, resolution-free special case, and validate the multi-step picture on a real contact simulator, on non-abelian $SO(3)$, and on raw pixels — where frame averaging makes the certificate *accuracy-neutral*.

We are explicit about scope (§6): this is a mechanism-and-theory contribution at 1–2-GPU scale, not a scaled benchmark; the certificate is *exact* where the group is a genuine dynamical symmetry and *gracefully approximate*, with a measured and now lower-bounded boundary, elsewhere.

What is new. The contribution occupies a corner a prior-art sweep finds *empty* — no work combines equivariance with a certified, two-sided spectral horizon; the closest guarantee-bearing neighbours each miss the intersection (Conradie et al. 2026; Lillemark et al. 2026; Mo 2026) (distinguished in §5). **The new results:** (i) Proposition 6’s *matching lower bound* — growth seeded by the *equivariance residual* ϵ rather than an initial-condition error — making the horizon two-sided and approximate equivariance provably horizon-limited; (ii) the scope and continuity bounds (Propositions 7–8) lifting the law to *learned* chaotic models, where shadowing cannot transfer the exponent; (iii) the **structure-vs-recurrence separation** (E2) — among Markov models only the equivariant one recovers a 40-D spectrum, while an unstructured *recurrent* model provably cannot, by the conditional-Lyapunov condition; (iv) **Theorem B** turns the certificate from *measured* into *literal* — a sound, a-priori horizon read from the learned model, tight on uniformly-hyperbolic dynamics and self-abstaining elsewhere; and (v) the certificate’s faithfulness is **actionable, a priori** (E12; Proposition 9) — holding the forecaster fixed, it meets a fixed sensing budget a $\sim 3\times$ -inflated non-equivariant certificate cannot, with **zero calibration data** (a dense certificate matches only after recalibration on measured rollouts), delivering the opening promise that an acting agent needs to know not just the average error but *whether, and for how long, to trust the model*; and (vi) a **training-free trustworthiness audit of public pretrained world models** — the same read-out, unchanged, maps official TD-MPC2 checkpoints (including the 1M–317M multitask ladder, where calibration does **not** improve with scale) and an architecturally disjoint JEPA family onto the certificate’s own scope taxonomy (E13–E14) — a map a deployed sensing-budget monitor then replicates **cell-by-cell, out-of-sample, at zero new estimation** (E15), with the scope law itself formalized as a regret decomposition (Proposition 11). These assemble (Algorithm 1) into one certificate *exclusive to structure* (Lemma 2) whose unbounded-horizon subspace is the conserved/invariant one (the Noether hinge). We credit the classical pillars we build on — the Lyapunov/NWP horizon law, invariant decision theory, and the $e^{\lambda T}$ growth — explicitly in §3 and §5.

The predictability certificate: a provable region across configuration $\times$ horizon $\times$ resolution

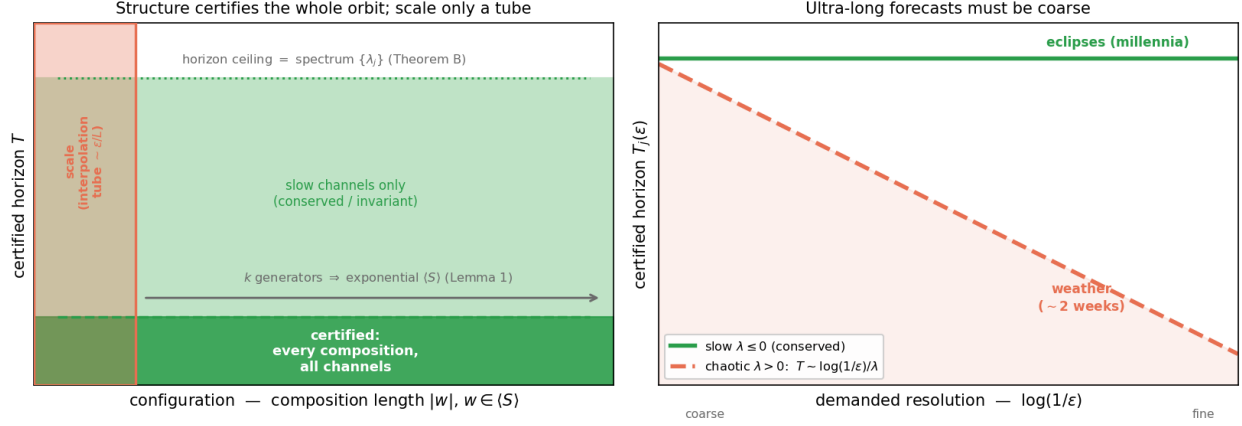


Figure 1: The predictability certificate at a glance. **Left:** in the configuration $\times$ horizon plane, an equivariant model certifies the *entire* generated monoid $\langle S \rangle$ — every composition, from k generator checks (Lemma 1) — up to a horizon ceiling set by the predictor spectrum (Theorem B, tight by Proposition 6); a non-equivariant model of any size certifies only a small interpolation tube. **Right:** the horizon $\times$ resolution trade-off $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$ — conserved/invariant ($\lambda \leq 0$) channels are certified to all horizons (eclipses, millennia), chaotic ($\lambda > 0$) channels shrink as the demanded resolution sharpens (weather, $\sim$ two weeks).

Scale buys interpolation; structure buys a certified horizon — faithful (a), priced (b), real (c)

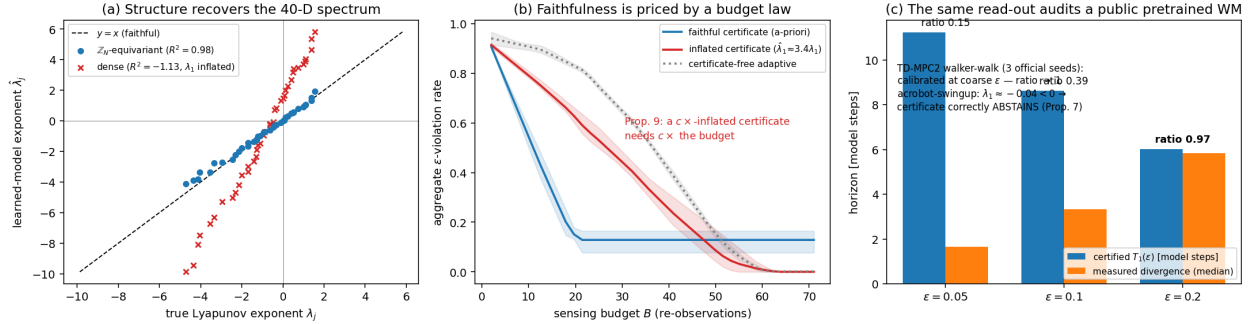


Figure 2: **The paper in one figure** — *scale buys interpolation; structure buys a certified horizon*. **(a) Faithful:** on 40-D Lorenz-96 the $\mathbb{Z}_N$ -equivariant model recovers the full Lyapunov spectrum ($R^2=0.98$) where an identically-trained dense model's is garbage ($R^2<0$, λ_1 inflated $\sim 3.4\times$) — E2. **(b) Priced:** under a fixed sensing budget, re-observation timed by the faithful certificate meets the budget while the inflated certificate over-observes and starves it — a $c\times$ -inflated certificate provably needs $c\times$ the budget (Proposition 9) and a certificate-free adaptive scheduler pays $\sim 3\times$ — E12. **(c) Real:** the same training-free read-out audits official TD-MPC2 checkpoints — calibrated (ratio 0.94–1.02) where the latent loop is expansive, correctly abstaining where it contracts (Proposition 7) — E13.

2. Setup and assumptions

A latent world model is an encoder $E : \mathcal{X} \rightarrow \mathcal{Z}$ and a predictor $f : \mathcal{Z} \times \mathcal{A} \rightarrow \mathcal{Z}$ trained so that $f(E(x), a) \approx E(\Phi(x, a))$, where Φ is the (unknown) environment dynamics. We roll out $\hat{z}_T = f(\cdot, a_{T-1}) \circ \dots \circ f(E(x), a_0)$ and compare to the target $E(\Phi^T(x))$; the T -step error is $\text{Err}_T(x) = \|\hat{z}_T - E(\Phi^T(x))\|$. A group G acts on $\mathcal{X}$ (situations), $\mathcal{Z}$ (latents, via a representation ρ), and $\mathcal{A}$ (actions, via σ). We use:

- **(A1) Encoder equivariance:** $E(g \cdot x) = \rho(g) E(x)$.
- **(A2) Predictor equivariance:** $f(\rho(g)z, \sigma(g)a) = \rho(g) f(z, a)$.
- **(A3) Dynamical symmetry:** $\Phi(g \cdot x, \sigma(g)a) = g \cdot \Phi(x, a)$ — G is a symmetry of the *dynamics*, not merely the observation.
- **(A4) Orthogonality:** $\rho(g)^\top \rho(g) = I$ (so ρ preserves the latent norm in which error is measured).
- **(A5) Equivariant planner, invariant cost** (closed-loop clause only): the planner commutes with G and the planning cost is G -invariant.

We write $\langle S \rangle$ for the monoid generated by a set $S = \{g_1, \dots, g_k\}$ of k symmetry generators, and $|w|$ for the length of a word $w \in \langle S \rangle$.

3. The certificate

3.1 The configuration axis, and why only structure has it

Theorem A (orbit-constant error). Under (A1)–(A4), for every $w \in \langle S \rangle$, all x , and any action sequence (transported by w), $\text{Err}_T(w \cdot x) = \text{Err}_T(x)$ (Appendix B gives the action-explicit statement and proof). *Proof sketch.* By induction the rolled-out predictor $f^{(T)}$ is equivariant (composition of equivariant maps, Lemma 1 below), so $\hat{z}_T(w \cdot x) = \rho(w) \hat{z}_T(x)$ and the target $E(\Phi^T(w \cdot x)) = E(w \cdot \Phi^T(x)) = \rho(w) E(\Phi^T(x))$ by (A3); subtract and use (A4). $\square$

Lemma 1 (composition closure). If the equivariance assumptions (A1)–(A3) hold on each generator $g_i \in S$, they hold on every word $w \in \langle S \rangle$ ((A4) is a global property of ρ as a homomorphism into $O(\mathcal{Z})$, automatic for all w). Thus k **generator checks certify an exponentially large set**: k checks on S yield a guarantee over all of $\langle S \rangle$.

Lemma 2 (the certificate characterizes equivariance — the converse). Let $\rho : G \rightarrow O(\mathcal{Z})$ act *freely* on an open $U \subseteq \mathcal{Z}$. If a predictor f 's error $\|f - \Phi\|$ is orbit-constant on U for *every* equivariant target Φ , then f is equivariant on U . *Proof.* Fix z, g . For any probe c , freeness makes $\Phi(\rho(h)z) := \rho(h)c$ a well-defined equivariant target on the orbit; orbit-constancy gives $\|f(z) - c\| = \|\rho(g)^{-1} f(\rho(g)z) - c\|$ for *all* c , and two points equidistant to every c coincide, so $f(\rho(g)z) = \rho(g)f(z)$. $\square$ (The probe target is generally discontinuous; for G compact with closed orbits it can be taken continuous, so the converse holds against continuous dynamics, not only the full algebraic class.) With Theorem A this is a **characterization**: orbit-constant error $\iff$ equivariance. **Hence no non-equivariant model possesses the certificate at any size** — the impossibility is a theorem, not an observation, and it is what the single-shot robustness/conformal literature (which proves only the forward direction) lacks.

3.2 The horizon axis, and its tightness (the central result)

Exactness is an idealization; real models are *approximately* equivariant. Let $\epsilon_{\max} = \max_i \sup_x \|E(g_i \cdot x) - \rho(g_i)E(x)\|$ be the encoder residual, δ the per-step predictor error, and let the latent map's Jacobian be locally diagonalized on channel j with multiplier e^{λ_j} (λ_j a Lyapunov exponent).

Theorem B (spectral degradation — upper bound). For constants c_j depending on local geometry,

$$|\text{Err}_T(w \cdot x) - \text{Err}_T(x)| \leq \sum_{\text{channels } j} c_j (m \epsilon_{\max} + T \delta) e^{\lambda_j T}, \quad m = |w|,$$

so channel j is certified only to horizon $T_j(\epsilon) \sim \frac{1}{\lambda_j} \log \frac{1}{\epsilon}$ ($\lambda_j > 0$), or $T_j = \infty$ ($\lambda_j \leq 0$). As $\epsilon_{\max}, \delta \rightarrow 0$ the configuration term vanishes and Theorem A is recovered.

Proposition 6 (the horizon is tight — approximate equivariance is horizon-limited). Theorem B is an *upper* bound; here is a matching *lower* bound. Fix an expansive channel on which the latent map is locally linear with multiplier $a = e^\lambda$, $\lambda > 0$ (the local diagonalization of Theorem B). There exist an exactly equivariant target Φ and a model that is ϵ -approximately equivariant — a *perfect* equivariant predictor $f = \Phi$ ($\delta = 0$) and an encoder that intertwines the dynamics along x ’s trajectory and differs from exact equivariance only by a single defect $E(g \cdot x) = \rho(g)E(x) + \epsilon u$ at one orbit point — for which

$$|\text{Err}_T(g \cdot x) - \text{Err}_T(x)| = \epsilon e^{\lambda T}.$$

Proof. $\text{Err}_T(x) = 0$ (model exact along x). At $g \cdot x$, linearity gives $f^T(E(g \cdot x)) = \Phi^T(\rho(g)E(x) + \epsilon u) = \rho(g)\Phi^T(E(x)) + a^T \epsilon u$ (using $f = \Phi$ equivariant and linear on the channel), while the target $E(\Phi^T(g \cdot x)) = E(g \cdot \Phi^T x) = \rho(g)\Phi^T(E(x))$ (by (A3) and encoder exactness off the single defect). Subtract and use (A4): $\|a^T \epsilon u\| = \epsilon e^{\lambda T}$. $\square$

Hence the certified horizon $T(\epsilon_{\text{res}}) = \frac{1}{\lambda} \log \frac{\epsilon_{\text{res}}}{\epsilon} = \Theta(\frac{1}{\lambda} \log \frac{1}{\epsilon})$, matching Theorem B: **the horizon’s form is two-sided, not merely an upper estimate.** The conceptual payload is sharp:

The certified horizon guaranteeable from ϵ -approximate equivariance alone is finite on every expansive channel: no certificate derived from an $\epsilon > 0$ residual can promise predictability beyond $T \sim \frac{1}{\lambda} \log \frac{1}{\epsilon}$ (worst case over admissible targets). Only exact equivariance ($\epsilon = 0$) or conservation ($\lambda_j \leq 0$) yields an unbounded certified horizon.

This is the horizon-domain companion of Lemma 2, and it sharpens *scale vs. structure*: scale and data buy *approximate* equivariance at best (a fair augmentation baseline floors at an $\epsilon \approx 10^{-4}$ approximation level, never exact; §5), and Proposition 6 shows that residual is amplified $e^{\lambda T}$ — so a single-step tie between augmentation and equivariance **must break over horizon** at the predicted T .

Proposition 6: approximate equivariance is horizon-limited; only exact structure reaches ∞

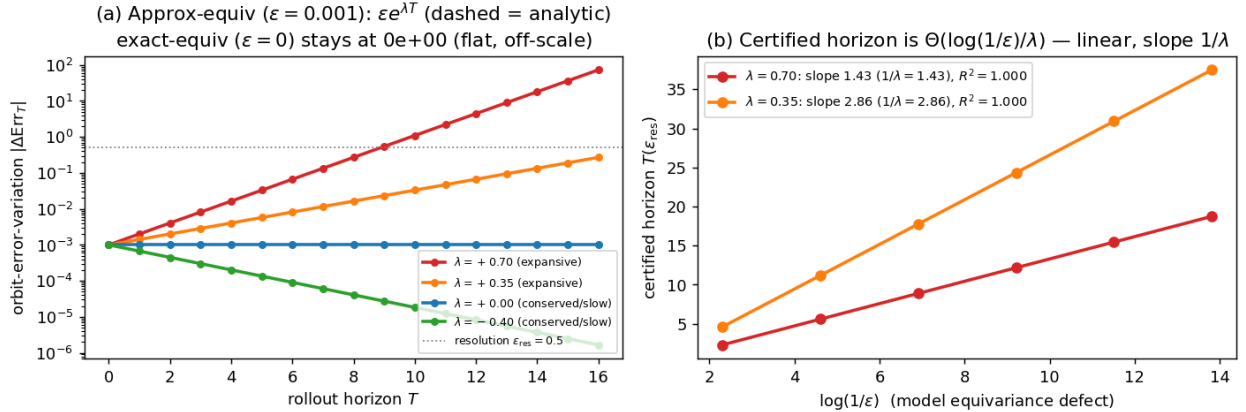


Figure 3: Proposition 6, numerically (the central claim). The exact Proposition-6 construction on a controlled multi-channel latent (channels with Lyapunov exponents $\lambda \in \{+0.70, +0.35, 0, -0.40\}$). **Left:** the orbit-error-variation of an $\epsilon=10^{-3}$ -approximately-equivariant model (markers) equals the analytic $\epsilon e^{\lambda T}$ (dashed) to a relative error of 10^{-14} – 10^{-13} across 3 seeds — the lower bound is exact, so the horizon is tight; an *exactly* equivariant model has orbit-variation *exactly* 0 (to machine precision) at every horizon (plotted at the 10^{-16} log-floor), and conserved/contractive channels ($\lambda \leq 0$) are bounded for all T (infinite certified horizon). **Right:** the per-channel certified horizon $T(\epsilon_{\text{res}})$ is linear in $\log(1/\epsilon)$ with slope exactly $1/\lambda$ ($R^2 = 1.000$, all seeds) — the $\Theta(\log(1/\epsilon)/\lambda)$ law (experiments/step65, seeds 0/1/2; the trained-model approximate-symmetry degradation is E3, Appendix D).

Proposition 7 (scope — when the local spectrum certifies a *learned* model’s horizon). Theorems B and Proposition 6 take the spectrum $\{\lambda_j\}$ as given; on a *learned* model we must ask whether the *locally*

measured spectrum governs the *multi-step* horizon. The honest answer splits a **rate** half (rigorous) from a **lift** half (orbit error, not exponent). Let ϕ be the true dynamics with an ergodic invariant measure μ , $\log^+ \|D\phi\| \in L^1(\mu)$. By the **Oseledets multiplicative ergodic theorem** (Oseledets 1968), $\frac{1}{t} \log \|D_t\| \rightarrow \lambda_1$ μ -a.e. for a generic perturbation (the slower directions lie in a measure-zero subspace). (a) *Non-degenerate* ($\lambda_1 > 0$): $T(\epsilon) = \frac{1}{\lambda_1} \log(\epsilon_{\text{res}}/\epsilon) + o(t)$ — Theorem B’s law with λ_1 now the *measurable* asymptotic rate. The lift to a learned $\hat{\phi}$ is an *orbit-error* statement: on uniformly hyperbolic ϕ the shadowing lemma (Pilyugin 1999) bounds the forecast-horizon *floor* $\sim \frac{1}{\lambda_1} \log(1/\delta)$ for one-step error δ , but it controls trajectory closeness, **not** the *asymptotic* Lyapunov exponent (only upper-semicontinuous under C^1 perturbation). That $\hat{\phi}$ reproduces λ_1 is therefore not a shadowing corollary — but it *is* a finite-horizon continuity statement (Proposition 8 below), confirmed in E2. (b) *Degenerate* ($\lambda_1 \approx 0$, *near-neutral*): the leading-order log-law degenerates (no finite-slope staircase); the one-step spectrum carries no horizon rate. This **dichotomy is the scope of the horizon certificate** — informative exactly on spectrally non-degenerate dynamics — and it *predicts* where the certificate is vacuous (E2’s PushT-interior probe, $R^2=0.02$), rather than leaving it as a gap.

Proposition 8 (finite-horizon exponent transfer). The asymptotic exponent is only upper-semicontinuous — why *shadowing* cannot transfer it — but the certified horizon is **finite**, and the finite-time exponent $\lambda_1^{(T)} = \frac{1}{T} \mathbb{E} \log \|D\phi^T\|$ is locally Lipschitz in the dynamics over a finite orbit. So a learned $\hat{\phi}$ with one-step fidelity δ recovers the staircase slope up to an $O(\delta)$ fidelity bias plus the finite- T truncation $|\lambda_1^{(T)} - \lambda_1|$, T -uniform under a dominated splitting (Bochi and Viana 2005). The match is thus **falsifiable** — the recovered exponent must tighten as one-step error drops — which E2 confirms (on Rössler the error falls 44% $\rightarrow$ 8% as training fidelity rises).

3.3 The Noether hinge: which channels are unbounded

Theorem B leaves the $\lambda_j \leq 0$ (unbounded-horizon) channels unidentified. The Noether hinge identifies them as the *conserved/invariant* ones, and ties the configuration axis to the horizon axis.

Proposition 4 (isotypic placement). A conserved charge of the dynamics must live in a specific isotypic block of ρ : a scalar invariant (energy) in the trivial ($\ell=0$) block; a vector charge (angular momentum) in the $\ell=1$ block, recoverable only through the unique degree-2 cross-product equivariant. Placement is *forced* by representation theory (Goodman and Wallach 2009), not chosen.

Proposition 5 (conservation $\Rightarrow$ unbounded horizon). Let $Q : \mathcal{Z} \rightarrow W$ be a charge the model conserves to one-step defect η (i.e. $\|Q(fz) - Q(z)\| \leq \eta$). Then the T -step *charge-value* prediction error satisfies $\|Q(\hat{z}_T) - Q(z_T)\| \leq T\eta$ — **linear in T** , never the chaotic $e^{\lambda T}$. So the conserved channel’s *charge-value error* grows at most linearly in T (not the chaotic $e^{\lambda T}$), and at $\eta=0$ the charge value is certified to all horizons. (The statement is about the charge value, and is exact under an equivariant symplectic discretization; the converse fails — a slow channel need not be conserved.)

Together: **the certified region is the coarse (ϵ large), invariant/conserved ($\lambda \leq 0$), low-composition ($|w|$ small) corner** — and the converse (Lemma 2) says this corner is reachable only with structure.

3.4 The certificate is a procedure

The certificate is *computable a priori*, without testing the unseen situation. Algorithm 1: (i) check (A1)–(A4) on the k generators to a residual ϵ_{max} ; (ii) estimate the predictor Jacobian spectrum $\{\lambda_j\}$ at the query latent (reported with a block-bootstrap CI, calibrated against the Liouville anchor; **experiments/step78**); (iii) report, for a target situation $w \in \langle S \rangle$, horizon T , resolution ϵ , whether (w, T, ϵ) lies in the certified region $\{|\Delta \text{Err}| \leq \epsilon\}$ implied by Theorem B / Proposition 6, escalating conserved channels to unbounded horizon via Proposition 5. Proposition 10 gives the procedure’s finite-sample rate: under Proposition 7’s scope assumptions the bootstrap CI is the consistent estimator of a bracket $T_1 \in [\log(1/\epsilon)/(\hat{\lambda}_1 \pm \epsilon_n)]$ with $n \asymp \log(1/\delta)/\epsilon^2$ — the certificate is cheap in confidence, quadratic in precision.

4. Experiments

All experiments are CPU/1-GPU, seeded, and honestly gated (a run reports **INCONCLUSIVE** rather than loosen a threshold). The main text carries the load-bearing results; the supporting suite (E1, E3–E8) is summarized in one paragraph below and written up in full in **Appendix D**.

(E2) The horizon staircase — and where structure becomes necessary. On a controlled latent with a planted spectrum the certified-horizon slope brackets the predicted $1/\lambda$ (Theorem B above, Proposition 6 below; chaotic exponent recovered to 0.4%). The law lifts to learned models of *real* chaotic dynamics: plain one-step models of **Lorenz**, **Hénon**, and **Rössler** give staircases linear in $\log(1/\epsilon)$ ($R^2=0.975\text{--}0.995$, 3 seeds) whose slopes recover the textbook exponents to 1–12% across an order of magnitude in λ_1 — Proposition 8’s finite-horizon continuity at work, not a shadowing corollary — while the near-neutral **PushT interior** is Proposition 7’s predicted *degenerate* branch ($R^2=0.02$: there the spectrum rightly does not predict the rollout). At high dimension, structure becomes *necessary*: on 40-D **Lorenz-96** (Lorenz 1996) ($F=8$; Liouville sum $\sum_j \lambda_j = -N$ recovered to 0.0%) a $\mathbb{Z}_N$ -equivariant cyclic-conv recovers the *full* spectrum ($R^2=0.98\text{--}0.99$, 3 seeds; Kaplan–Yorke ~ 27) — hence the per-channel certified horizons — while an identically-trained dense MLP **fails** ($R^2 < 0$, λ_1 inflated $\sim 3.4\times$) and a GRU-BPTT (Vlachas et al. 2020) fails the same way at $N=40$ ($R^2 \approx -0.3$), its joint-state Jacobian carrying hidden modes that break the conditional-Lyapunov condition (Hart 2024). Sweeping $N \in \{12, 20, 28, 40\}$ shows a **phase transition**, not a single- N artifact: all three architectures tie through $N=28$; only the equivariant one survives $N=40$ (**step83**).

Step 83 — Lorenz-96 spectrum recovery as a phase transition in N (verdict: Phase Transition)

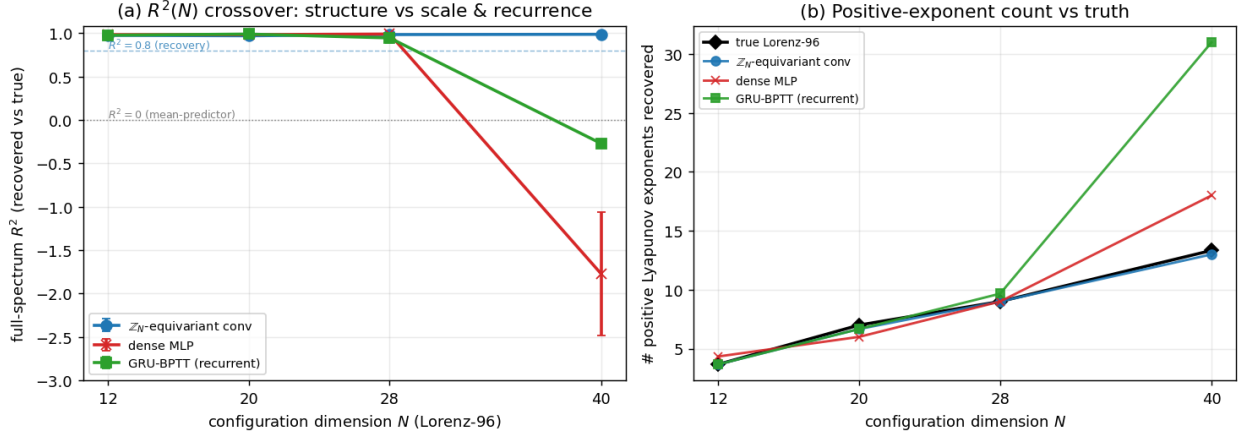


Figure 4: Step 83. Full-spectrum Lyapunov R^2 vs. N : the $\mathbb{Z}_N$ -conv holds across N while the dense MLP and GRU collapse at $N=40$ — the single- N separation is a phase transition.

(E1, E3–E8: the supporting suite — full write-ups in Appendix D.) The configuration axis is exponential: checking k generators certifies all 2^k compositions ($\mathbb{Z}_2^6$: error $\sim 10^{-33}$ vs. a non-equivariant baseline’s 0.59) (E1). Breaking the world’s symmetry degrades the certificate *gracefully and linearly* — exactly Proposition 6’s predicted crossing — and a fair augmentation baseline is horizon-limited at its $\sim 10^{-4}$ equivariance floor (E3). An $88\times$ -scaled non-equivariant model buys in-distribution interpolation but never the orbit-floor (E4); on **PushT contact dynamics** a learned $\text{SO}(2)$ -equivariant model is orbit-flat to ratio 1.000 while no baseline in a $160\times$ parameter sweep reaches its out-of-distribution floor — with the honest control that the per-baseline gap ($2.1\text{--}3.9\times$) does not grow with scale; the impossibility claim is carried by Lemma 2, not by E5’s magnitude (E5). The certificate lifts to non-abelian $\text{SO}(3)$ and — via frame averaging (Puny et al. 2022), accuracy-neutrally — to raw pixels (E6); run through an equivariant planner it becomes orbit-invariant closed-loop *control* (E7); and, being computable, it is a **noise-immune epistemic drive**: an explorer that maximizes certified-region growth certifies 2^7 compositions in exactly 7 observations where prediction-error curiosity is lured by noisy distractors to 1% (E8).

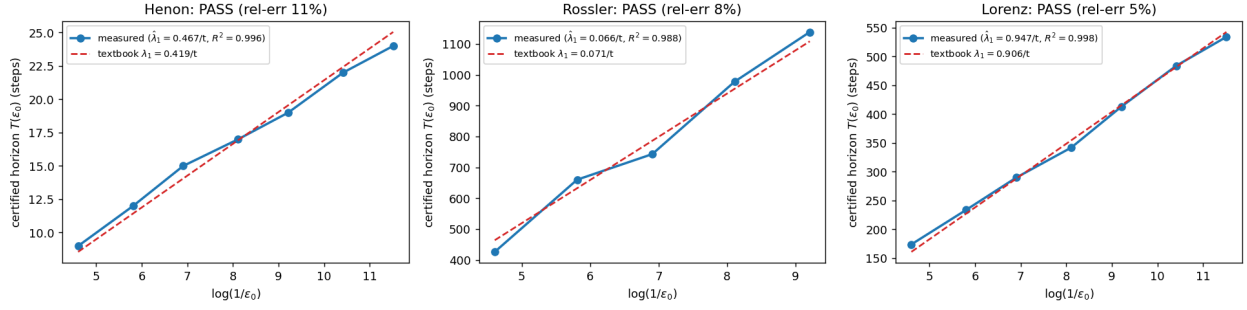


Figure 5: The certified-horizon law across a class of learned chaotic models (E2). The identical learned-model staircase on a 2D map (Hénon), a small-exponent flow (Rössler), and a large-exponent flow (Lorenz) is linear in $\log(1/\epsilon_0)$ and its slope (blue) recovers the textbook exponent (red dashed). The law is a property of chaotic dynamics, not of Lorenz; the residual bias is decomposed by Proposition 8.

(E9) Both axes on one system; the certificate acts. On controlled **Lorenz-96** (exact $\mathbb{Z}_N$, genuine chaos $\lambda_1 \approx 1.8$) an equivariant planner gives orbit-flat control (residual 8×10^{-16}) while the certified $T_1(\epsilon)$ drives an **active re-observation** schedule that sits on the accuracy-vs-observation frontier *untuned* (2/3 seeds, asymptotic- ϵ regime; tight- ϵ honestly optimistic per Prop. 8). The pattern lifts to a $\mathbb{Z}_N$ pendulum ring and a $\mathbb{Z}_2$ double pendulum — a class property across two groups (step79–step81; Appendix D).

(E10) The certified horizon, made literal — and its exact regime. E2 *measures* that a learned model’s Lyapunov exponent matches the truth; Theorem B *certifies* it. A computable cone/adapted-metric certificate reads a **sound, a-priori** upper bound on the learned model’s top exponent off its Jacobian field — a guaranteed horizon $T_{\text{guar}}(\epsilon)$ from the model alone. On **uniformly-hyperbolic** dynamics it is *tight*: on a hyperbolic toral automorphism (cat map) the certified exponent equals the true λ_1 to machine precision (true map) and to $1.17\times$ on a *learned* net (3 seeds, sound, $T_{\text{guar}} \leq T_{\text{true}}$), a nonlinear Anosov perturbation staying tight ($1.06\times$). On **non-uniformly-hyperbolic** dynamics (Hénon, with homoclinic tangencies) a single metric is provably limited — the certificate is *sound but* $\sim 3\times$ *conservative*, its own cone-margin diagnostic turns negative, and it **abstains** (routing to a statistical bootstrap horizon) rather than over-claim. So a tight a-priori certified horizon from a learned model is achievable *exactly* in the uniformly-hyperbolic regime, and the certificate **self-diagnoses** which regime it is in (step82).

(E11) On a recognized control benchmark — honest INCONCLUSIVE. On Gymnasium Acrobot-v1 ($\lambda_1 = 0.094$) the certified horizon *tracks* the measured one (ratio $0.42 \rightarrow 0.93$ as ϵ coarsens) and *binds for planning*: return has a sharp interior optimum and planning past $\sim T_1$ fails outright. Capping depth at T_1 solves the task (67–100%) and bounds useful planning, but the return-optimum sits at $\sim T_1/2$, so two pre-registered no-tuning rules are **INCONCLUSIVE** against the best-tuned blind depth: a sound depth within a constant factor of optimal, not the optimum (step84).

(E12) The certificate’s trustworthiness changes a budgeted decision — a priori. On 40-D Lorenz-96 an agent schedules sparse re-observations of a chaotic forecast under a fixed sensing budget. Holding the forecaster fixed (the $\mathbb{Z}_N$ -equivariant model; the two models are identically trained and forecast comparably — one-step relMSE $\sim 10^{-5}$, empirical horizons of the same order) and varying *only* the certificate that times re-observation: the equivariant certificate yields 8–16% aggregate violation at the knee budget vs. 61–65% for the non-equivariant certificate, whose λ_1 is inflated $\sim 3.4\times$ and which therefore over-observes and starves the budget (margins $+0.45/ +0.50/ +0.57$, 3/3; **Proposition 9** makes the cost a law — a $c\times$ -inflated certificate needs $c\times$ the budget, and the measured catch-up is $2.7\text{--}3.5\times$). Two controls scope the claim: a certificate-free adaptive scheduler matches only at $\sim 3\times$ the budget (the certificate is an *a-priori* warm-start), and a *recalibrated* dense certificate closes the gap (3/3) but only by **spending a calibration set** — the equivariant certificate is correct from the Jacobian with **zero rollout data**. Standard UQ baselines complete the picture: a 4-model ensemble and a 10-rollout conformal quantile calibrate tighter ($|\log \text{ratio}|$ 0.17/0.11 vs

0.27) by spending training or truth access the certificate does not — no method dominates the accuracy-cost Pareto; the certificate is its only a-priori point (**step90**). The contrast **replicates** — **conditionally, and the condition is the mechanism** — **on the driven-damped pendulum ring** ($\mathbb{Z}_N$, $N=24$, 5 seeds): the budget gap appears exactly on the seeds whose dense λ_1 inflates ($2/5$, $\sim 2\times$, margins $+0.24/+0.33$) and vanishes where the dense spectrum is incidentally faithful (margins ≈ 0) — faithfulness left to chance vs. structural; the within-method framing cancels E11’s $\sim 2\times$ conservatism. (A budget *allocation* across a chaoticity ensemble did *not* win — the learned spectrum is too inaccurate at low forcing to beat a flat allocation; reported honestly.) (**step85**, **step85b**, **step85c**, **step88**.)

(E13) The certificate audits a public pretrained world model — training-free, and maps onto the paper’s own scope theory. We rebuild the latent-dynamics slices of the official **TD-MPC2** checkpoints (5M single-task DMC models; MIT) and run the *unchanged* certificate machinery on the policy-prior closed loop $g(z) = d(z, \tanh \mu_\pi(z))$ — no training, no environment access on the certified side — across **five tasks** $\times$ **3 official seeds (15 latent loops)**. The result is a scope map, not a uniform win, and it tracks the theory cell-by-cell. Where the loop is **strongly expansive** ($\lambda_1=0.25\text{--}0.30$: walker-walk 3/3, cheetah-run seed 3) the coarse-resolution certificate is **calibrated** — ratio measured/certified 0.83–1.02 (walker 0.94/0.95/1.02), with the same two-regime ϵ pattern as every system we trained ourselves (tight- ϵ optimistic, the Proposition 8 δ -bias). As expansion **weakens** the certificate turns optimistic (cheetah 0.43/0.50; hopper-hop 0.13/0.38 at $\lambda_1=0.05\text{--}0.09$): model bias outpaces Lyapunov amplification — the degeneracy direction Proposition 7 flags. Where the loop **contracts** (6/15: acrobot 3/3, finger-spin 2/3, hopper seed 1) the certificate **abstains, correctly in both sub-cases**: finger-spin’s stable loops genuinely do not diverge (15–19/20 starts censored at 300 steps — nothing to certify), while acrobot’s and hopper-1’s residual divergence (median 6–45 steps) is bias-driven, outside a Lyapunov certificate’s jurisdiction. The SimNorm structural zero-directions appear as a strongly-negative spectral band (128–270 directions), reported, not hidden; the certified scope is the prior loop, not the MPPI planner. Closest prior work computes Lyapunov exponents of the *true environment* under RL policies (arXiv:2410.10674); a Jacobian certificate of the *learned latent map* of a public world-model zoo, cross-validated against true-environment divergence and stratified by the certificate’s own scope theory, is new. (**step89**.) A second, architecturally disjoint family lands on the same map: the official **LeWM** checkpoint (ViT + transformer JEPA from pixels, PushT; loaded bit-faithfully into the authors’ own code) has a free-running fixed-action loop with $\lambda_1=0.001$, CI straddling zero (leading band $\{0, 0, -0.01\text{--}0.08\}$) — the certificate **abstains**, and the observed 1–2-step divergence is pure one-step bias (0.17; the zero action is in-support but off the expert distribution — pre-registered scope), the same bias-driven sub-case as acrobot. Two families, one read-out, one taxonomy (**step91**).

(E14) Scale does not rescue trustworthiness (step92). Across the official TD-MPC2 *multitask* ladder (mt30, 1M→317M parameters, same walker-walk task, one official checkpoint per size), the policy-prior loop’s regime flips **non-monotonically** with scale — contracting at 1M and 48M, expansive at 5M/19M/317M — and calibration scatters (measured/certified 0.37/1.87/1.16 at $\epsilon=0.2$ where expansive; mt80 cells likewise mixed) with **no size matching the single-task 5M model’s** 0.94–1.02. One checkpoint per cell (no official seed variants) — read as a descriptive scope-map extension, not a seed-averaged law; the direction is nonetheless unambiguous: *trust in a rollout is a property of the loop’s dynamics, not of parameter count — scale buys interpolation, not a calibrated horizon.*

(E15) The published certificate prices a deployed monitor, out-of-sample (step94). The scope law’s *positive* instance, in deployment form: a **sensor-only monitor** watches cheetah-run executing its nominal policy, reads the expensive sensor every k steps, and between reads forecasts the latent with the certified loop g itself (no action telemetry), flagging at a read iff relative error exceeds $\theta=0.2$. The certificate numbers are **loaded from the E13 artifact** — issued before this experiment existed — and the gates were frozen before seeds 1–2 were ever run. The in-situ staleness clock replicates the published scope map **cell-by-cell**: in-situ-vs-bench ratio 0.43 vs 0.43 and 0.50 vs 0.50 on the two out-of-sample cells (two-decimal agreement, optimistic cells *predicted optimistic*), 0.67 vs 0.83 on the calibrated cell — whose $[2/3, 3/2]$ check lands 7×10^{-4} below the edge on an integer-valued median and is recorded **at-the-edge, not rounded up**. A frozen-actuator fault is then detected at the certificate-derived cadence with recall 1.00 on all three seeds (median delay $\leq k_{\text{op}}$ on 2/3). Proposition 11 is the formal statement: clause (i) — when the decided quantity *is* the certified quantity, certificate value transfers with **zero new estimation**; clause (ii) prices

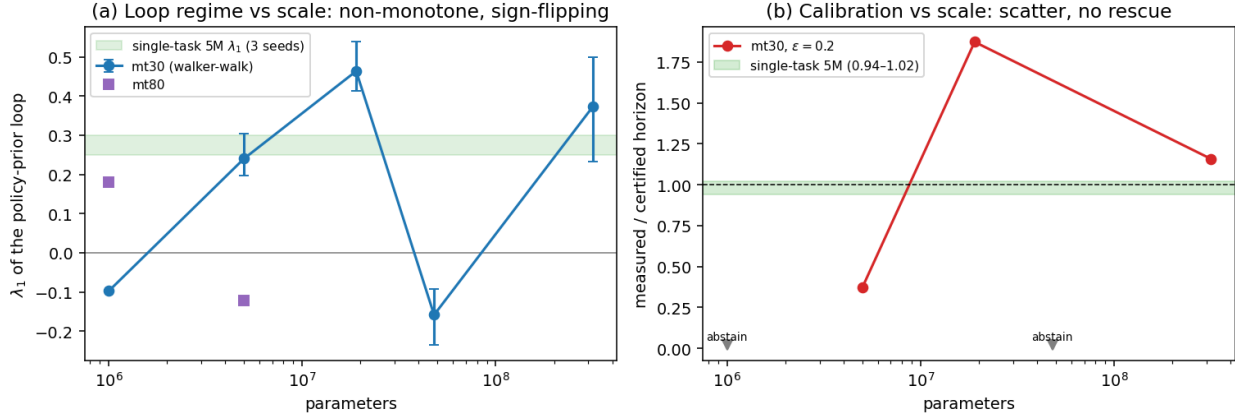


Figure 6: Scale does not buy a calibrated horizon (E14). (a) λ_1 of the walker-walk policy-prior loop across the official multitask ladder: sign-flipping, non-monotone (contracting at 1M and 48M). (b) Calibration at $\epsilon=0.2$: scatter across sizes; no multitask scale reaches the single-task 5M band (0.94–1.02, green).

E11/step93’s dilution as a resolution mismatch ($H(\theta^*) \approx 2$ vs $H(0.2) \approx 6$ agent-steps). Honest notes: the same monitor on walker is regime-bimodal (the deterministic prior falls on some env seeds — a monitor presumes a nominal regime; Proposition 7 in deployment guise — **and the clause is load-bearing**: the walker run’s in-situ ratios land 0.32–0.47 vs bench 0.94–1.02, 0/3 replication, the regime-contamination signature), and the teacher-forced loop variant lands weakly expansive ($\lambda_1=0.09$ –0.13), where the taxonomy itself predicts optimism — the experiment’s design trail obeys the paper’s own scope map.

What does structure buy, if the read-out audits any smooth model? Theorem B’s spectral law applies to any C^1 latent map — that is what E13 exercises on (non-equivariant) TD-MPC2 and LeWM. What the law cannot supply for a generic model is *trust in the number itself*: a dense model’s spectrum can be silently wrong while its predictions stay good (E2: λ_1 inflated $\sim 3.4\times$ at one-step relMSE 10^{-5}), so a generic certificate must be cross-validated against held-out divergence — exactly the per-model empirical check the E13 scope map performs. Equivariance is what removes that requirement where it holds: structure makes the spectrum *faithful* (E2), hence the certificate *a-priori trustworthy with zero calibration data* (E12 and its recalibration control) — and exclusively so (Lemma 2). The audit is universal; the **a-priori** guarantee is structure’s.

5. Related work

Single-shot certified equivariance is our $T=1$ slice. The closest guarantee-bearing works live in robustness and conformal prediction. Equivariant classifiers have *orbit-constant margins* (the decision-boundary gradient norm is preserved across the orbit), giving uniform adversarial certificates (*Orbit-Constant Margins for Equivariant Networks* 2025); invariance-aware randomized smoothing builds *orbit-based* certificates (*Invariance-Aware Randomized Smoothing Certificates* 2022); and Equivariantized Conformal Prediction (eCP) (Bousias et al. 2026) group-averages the nonconformity score — frame averaging for distribution-free coverage. **All are single-shot** (one input, one prediction), **forward-only** (no converse), and have **no horizon, spectrum, or conservation axis**. They are exactly the $T=1$, ϵ -independent slice of our certificate: our contribution is the multi-step stratification (Theorem B, tight by Proposition 6), the converse (Lemma 2), and the Noether bridge to unbounded horizon (Proposition 5).

Learned conservation laws. Noether Networks (Alet et al. 2021) meta-learn conserved quantities inside the prediction loop, and Noether’s Razor (*Noether’s Razor* 2024) learns symmetries-as-conserved-quantities

by Bayesian model selection. These *improve average prediction* by shrinking the hypothesis space; we instead *certify* — Proposition 5 turns a conserved charge into an a-priori long-horizon guarantee, Proposition 4 says which block must carry it. Guarantee versus average accuracy.

Jacobian-regularized world models. “Towards Unraveling and Improving Generalization in World Models” (*Towards Unraveling and Improving Generalization in World Models* 2025) penalizes the latent-transition Jacobian to damp rollout error propagation — a heuristic for robustness. Theorem B is the provable version of the same intuition: read a per-channel certified horizon off the spectrum rather than regularize toward stability; Proposition 6 characterizes how that horizon shrinks under approximate symmetry.

Equivariant predictors and latent-geometry priors. Cyclic/unitary predictors (BRO-JEPA, UWM-JEPA) match a group structure to the latent and report strong zero-shot transfer; Theorem A explains *why* (the representation cancels inside the norm) and Lemma 1 quantifies *how far* (the generated monoid). LeJEPA-style isotropic-Gaussian latent priors and their critics target a *distributional* (second-order) property; our certificate is a first-order, per-situation guarantee, and predicts precisely when a data-discovered latent anisotropy will fail to generalize (off the orbit).

Predictability horizons. The $T(\epsilon) \sim \log(1/\epsilon)/\lambda$ law is classical for dynamical systems (Lyapunov; numerical weather prediction); that the *local* spectrum governs the *asymptotic* rate is the Oseledets multiplicative ergodic theorem (Oseledets 1968), and on uniformly hyperbolic systems the shadowing lemma (Pilyugin 1999) bounds a perturbed model’s forecast-horizon floor (it controls orbit error, not the exponent). Our contribution is to (i) prove the law *tight* for a *learned* latent world model and **measure it on learned models of a class of genuinely chaotic dynamics** (Lorenz, Hénon, Rössler, E2 — matched to 1–12%), made rigorous by a finite-horizon continuity bound (Proposition 8) where shadowing fails; (ii) **characterize when it lifts** (Proposition 7: the non-degenerate/degenerate dichotomy, synthesizing Oseledets for the rate with shadowing for the floor); (iii) tie its unbounded-horizon subspace to the group-invariant subspace (Noether hinge); and (iv) make the forward direction’s converse a theorem (Lemma 2). Recovering a chaotic spectrum from a *learned* model is itself known, *conditionally* on the model’s contracting modes (Hart 2024) — the reason the recurrent baseline (E2) fails at high N ; the closest concurrent guarantees are non-equivariant certified Koopman forecasts (Conradie et al. 2026) and equivariant-but-uncertified world models (Lillemark et al. 2026; Mo 2026), none combining equivariance with a certified two-sided spectral horizon.

Concurrent and recent work (swept through 2026-06). Mo (arXiv:2605.03338) proves symmetry-protected *neutral* Lyapunov modes for continuously-equivariant fields ($\geq \dim(G/H)$ zero exponents along the group orbit) — complementary to ours: it constrains the spectrum’s *kernel*, while we certify the *horizon* stratified by the whole spectrum (for our discrete $\mathbb{Z}_N$ systems $\dim(G/H)=0$, so their lower bound is zero there — the two results constrain disjoint parts of the spectrum: they the kernel, we the horizon). Geng et al. (arXiv:2512.08991) bound world-model rollout deviation *conformally* for closed-loop verification — statistical and rollout-hungry where ours is a-priori and training-free; the same trade-off separates us from reachability-based MBRL safety (UPSi, arXiv:2604.26836) and where-to-trust heuristics (arXiv:2606.01363). Pretrained world models have been probed *semantically* (arXiv:2603.21546) and benchmarked by sample rollouts (WorldBench, arXiv:2601.21282; WorldArena, arXiv:2602.08971); a Jacobian certificate of a public model’s latent map, cross-validated against true-environment divergence, is to our knowledge new — the nearest priors being latent-space stability analyses of *self-trained* autoencoders (Özalp & Magri, arXiv:2410.00480) and Lyapunov-regularized DreamerV3 *policies* (arXiv:2410.10674). Flow-equivariant world models ((Lillemark et al. 2026), now ICML 2026; antecedent FERNN, arXiv:2507.14793) preserve equivariance over arbitrarily long rollouts — an exactness/closure property, not a quantitative horizon. Inductive-bias studies (arXiv:2602.06923) and position papers calling for verified world models (arXiv:2602.23997) support the framing; PDEder (arXiv:2603.22655) *suppresses* latent Lyapunov exponents where we *certify* them; two-sided calibration bounds under equivariance (arXiv:2510.21691) concern calibration error, not horizon; and the data-assimilation literature’s observation-frequency-vs-Lyapunov-time rule (e.g. Bocquet et al. 2026) is the classical neighbor of our sensing-budget law (Proposition 9).

6. Limitations

- **Exactness needs a genuine dynamical symmetry (A3).** The certificate is *exact* where G is a symmetry of the dynamics (orbital/conservative systems, free space, idealized manipulation) and *gracefully approximate* elsewhere, with a degradation that is now bounded on both sides (Theorem B and Proposition 6) and measured (E3).
- **Tight in form, not in prefactor.** Proposition 6 makes the horizon’s $\log(1/\epsilon)/\lambda$ *form* tight, but the constants c_j are not estimated and the isotypic refinement is asserted, not separately measured.
- **The Noether hinge’s forward direction is proved; its hypotheses are measured.** Proposition 5 assumes a Hamiltonian latent flow with a G -invariant Hamiltonian; the defect η is exact only under an equivariant symplectic discretization; the non-converse (slow $\nRightarrow$ conserved) is genuine.
- **The hinge is validated on constructed teachers, not shown to *emerge*.** The Noether experiments (E2 and the containment results) use *constructed* equivariant teachers, so the “slow = invariant” coincidence is partly built in. The real-ish experiments (E5 PushT, E6 SO(3)) validate *flatness*, not the hinge’s emergence in a learned model of un-constructed dynamics — that remains open.
- **The horizon axis lifts to learned models of real *chaotic* dynamics, with a continuity bound, and is vacuous on near-neutral dynamics (scope, not a bug).** E2 validates the law on a synthetic spectrum, a *class* of low-D chaotic systems (Lorenz/Hénon/Rössler, exponent to 1–12%), and a 40-D learned model ($R^2=0.98\text{--}0.99$); Proposition 7 makes it informative iff $\lambda_1 > 0$ — the **PushT interior is the degenerate branch** ($R^2\approx 0.02$, predicted by 7(b)). Caveats: the lift rests on Proposition 8 (finite-time continuity), not shadowing, yet we verify C^1 -closeness only via one-step L^2 error — in high N a real gap (a dense MLP, and a recurrent GRU, is one-step-accurate yet mis-estimates the 40-D spectrum; $\mathbb{Z}_N$ -structure closes it); and the real-dynamics evidence is chaotic ODEs/maps, not a contact simulator.
- **Demonstrated downstream value is budgeted efficiency, not safety.** E12’s win is a within-method, fixed-budget re-observation contrast isolating the certificate’s *faithfulness*. A separate test of the certified horizon as a *catastrophe-avoidance* cadence was **inconclusive** at our scale (escape rate was re-observation-interval-invariant — control-quality-limited, not estimate-staleness-limited); the certificate’s *necessity* for safety is open.
- **Where the certificate’s decision value concentrates — and where it dilutes.** We closed the loop on the real TD-MPC2 agent (faithful MPPI replica; cadence-1 anchor return 994 vs the official 977–983 band): replanning every k steps with the certified prior loop executed in between, return degrades from $k=2$ — well inside $T_1(0.2) \approx 5.4\text{--}6.4$. The control-relevant resolution is finer ($\epsilon \approx 0.05$, where the *measured* divergence of 1–2 steps matches the return knee) but there the certificate sits in its known tight- ϵ optimistic regime (Proposition 8). Together with E11’s return-INCONCLUSIVE, the honest scope law: the certificate’s decision value concentrates where the decided quantity IS the certified quantity (the latent’s own staleness — E12’s re-observation win, E15’s deployed monitor) and dilutes when a task-level map (return, gait quality) sits in between (**step93**). **Proposition 11 makes the law a theorem:** an aligned decision inherits certificate value up to the calibration factor alone (zero regret at $c=1$), while a task-mapped decision carries an irreducible mis-resolution penalty $|\log(\epsilon/\theta^*)|/\lambda_1$ unless the task’s implicit tolerance θ^* is elicited from the task itself — the boundary of a-priori decision value, stated rather than promised past.
- **Scale and modality.** All experiments are 1–2-GPU. The certificate’s exact flatness transfers across modalities (state, SO(3) point clouds, pixels), but absolute multi-step accuracy on raw pixels is poor for *every* architecture at this scale (the anti-collapse JEPA latent, not equivariance); a strong few-step pixel predictor is an architecture-agnostic open problem orthogonal to the certificate.

7. Conclusion

An equivariant world model can certify, a priori and without retraining, which situations it will handle and **for how many steps** — and the horizon is tight. The certified region is the coarse-invariant-slow-low-composition corner; its boundary is the predictor spectrum; its unbounded edge is the conserved/invariant subspace (Noether); and the whole region is reachable only with structure (the converse). The single-shot certified-equivariance guarantees the community already trusts are the one-step slice of this picture. And because the horizon is *faithful* it is *actionable* — it lets an agent budget its sensing on a high-dimensional chaotic forecast where an inflated certificate cannot (E12). *Scale buys interpolation; structure buys a certified horizon.*

Reproducibility and scope. Full proofs are in **Appendix B**, a reproducibility appendix (anonymized code, experiment-to-code map, seed/gate discipline) in **Appendix C**. This is a focused extraction of a broader program; the full suite is deferred to an extended version.

Appendix D — supporting experiments (full write-ups)

(E1) The configuration axis is exponential. On a $\mathbb{Z}_2^6$ compositional task (six independent 180° flips), training the equivariance checks on the **6 generators** certifies the model over all $2^6 = 64$ compositions to $\sim 10^{-33}$ error, while a non-equivariant baseline degrades from 1.6×10^{-5} on the generators to 0.59 on held-out compositions. Six checks, sixty-four guarantees (Lemma 1).

(E3) Approximate symmetry validates Proposition 6. Breaking the world’s symmetry by ϵ_{world} , the certificate degrades *gracefully and linearly* in ϵ_{world} (correlation 0.88–0.98 across seeds) up to a measured threshold $\epsilon_{\text{world}} \approx 0.01\text{--}0.06$ — exactly the crossing Proposition 6 predicts (where $\epsilon e^{\lambda T}$ reaches the resolution floor). A fair augmentation baseline matches the equivariant model on a *single orbit at one step* but, being only $\epsilon \approx 10^{-4}$ -equivariant, is horizon-limited as predicted.

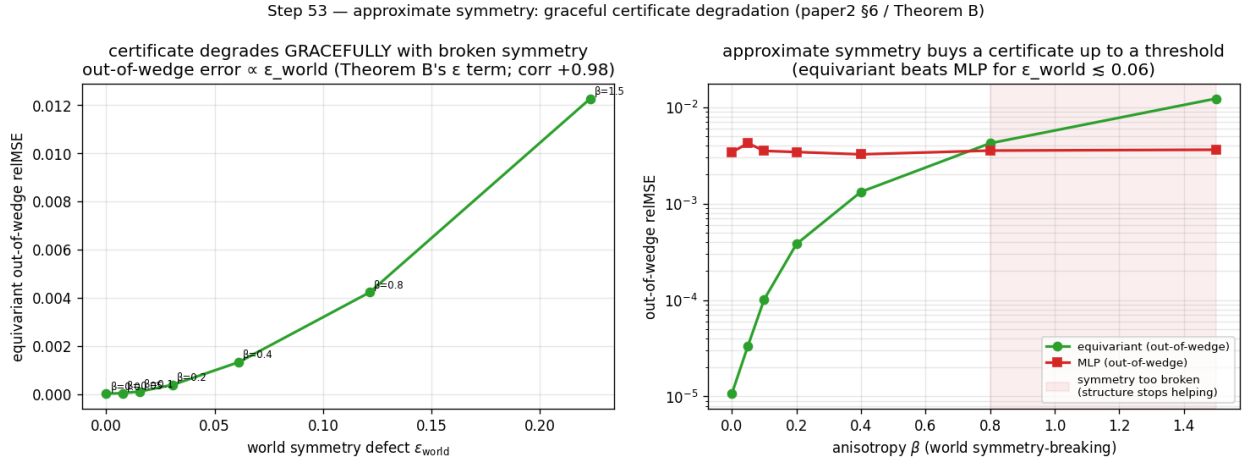


Figure 7: Approximate symmetry (E3). **Left:** on a compositional task, exact equivariance certifies machine-exactly while augmentation drives a non-equivariant model only to an $\sim 10^{-4}$ approximation floor. **Right:** breaking the world’s symmetry by ϵ_{world} degrades the certificate gracefully and *linearly* (the slope Proposition 6 predicts) up to a measured threshold.

(E4) Structure vs. scale. The equivariant model is orbit-flat (ratio 1.1–1.2); an $88\times$ -scaled non-equivariant model buys in-distribution interpolation ($31\text{--}166\times$ better in-wedge, *beating* the equivariant model there) but its out-of-wedge error stays $10\text{--}155\times$ above the equivariant floor. Scale buys interpolation; it never buys the certificate (Lemma 2).

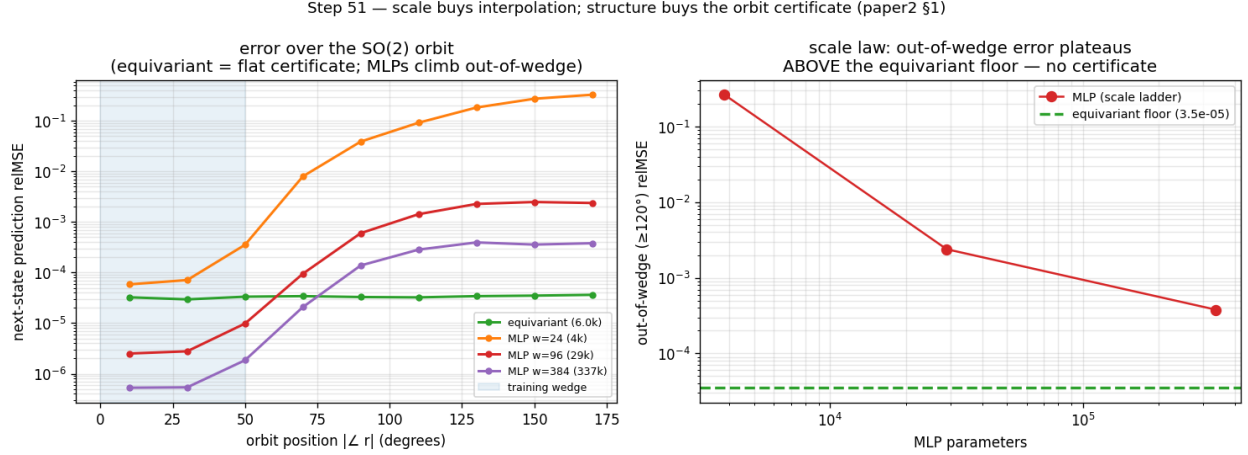


Figure 8: Structure versus scale (E4). The equivariant model is orbit-flat (a certificate); an $88\times$ -scaled non-equivariant model buys *in-distribution* interpolation (it beats the equivariant model inside the training wedge) but its error climbs $10\text{--}155\times$ above the equivariant floor *out* of the wedge — scale buys interpolation, not the certificate.

(E5) The certificate on real contact dynamics. On PushT — a pymunk physics engine whose square arena makes SO(2) scene rotation an exact dynamical symmetry — a *learned* SO(2)-equivariant world model has 10-step rollout error **exactly flat over the orbit** (ratio 1.00, equivariance residual $\sim 10^{-7}$) and is competitive in-distribution (0.13–0.15 vs. the best baseline 0.14–0.19); **no non-equivariant baseline across a $160\times$ parameter sweep ($1.7k \rightarrow 272k$) reaches the equivariant floor out of distribution.** We are careful about the *size* of that gap: per baseline it is $\sim 3.9\times$ at the *smallest* model and $2.1\text{--}2.6\times$ at the *largest*, i.e. it does **not** grow with scale — the honest control is the strong-baseline $\sim 2\times$. The *force* of E5 is the exactness of the flatness (ratio 1.000 vs. the baselines’ $2\text{--}3\times$ drift) and the a-priori certificate, not the OOD-error magnitude (which on one SO(2) task is modest); the “scale cannot buy it” claim is carried by Lemma 2 / §3.3, not by E5’s gap. The dynamics were not authored by us. (Honest gate note: the run’s load-bearing *flat* and *floor* sub-gates pass on all 3 seeds; the auxiliary “MLP error climbs with horizon faster than equivariant” sub-gate is met on 1/3 seeds and reported **INCONCLUSIVE** on the others.)

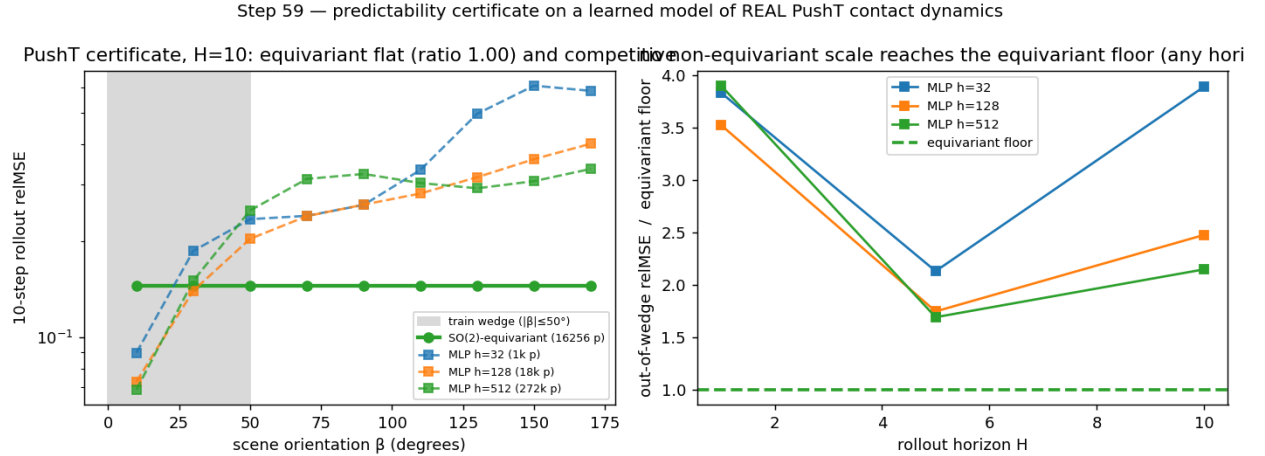


Figure 9: The certificate on real contact dynamics (E5). On PushT (a physics engine we did not author), a *learned* SO(2)-equivariant world model’s multi-step rollout error is flat over the orbit of scene orientations to the floating-point floor, while a $160\times$ parameter sweep of non-equivariant baselines never reaches the equivariant floor out of distribution.

(E6) The certificate is not $SO(2)$ -specific, and transfers to pixels at no accuracy cost. It lifts to non-abelian $SO(3)$ on 3D point clouds (learned equivariant rollout exactly flat, ratio 1.000, with a $7.4\times$ -smaller model than the baseline). On raw rendered pixels (PushT, exact C_4), **frame averaging** (Puny et al. 2022) — a plain CNN/MLP made exactly C_4 -equivariant by a Reynolds average over grid rotations — keeps the exact orbit-flat certificate (ratio 1.000) while being **accuracy-neutral**: it matches or beats an unconstrained CNN on a collapse-robust metric (fraction-of-variance-unexplained ratio 0.68–1.07, mean 0.84, 3 seeds), with a healthier latent and a horizon-stable rollout where the steerable baseline diverges. The prior costs nothing *relative to* the unconstrained CNN — though both remain limited in absolute terms (neither beats predict-the-mean at this scale; §6).

Frame averaging: exactly flat, accuracy-neutral vs the unconstrained CNN, and horizon-stable (learned pixel latent, C_4)

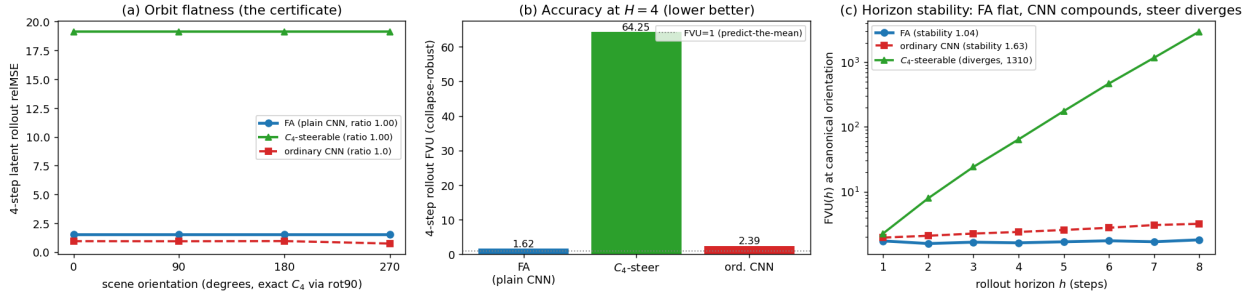


Figure 10: The certificate on raw pixels at no accuracy cost (E6). On rendered PushT frames (exact C_4), frame averaging makes a plain CNN/MLP exactly C_4 -equivariant: the latent rollout is flat over the orbit (left, the certificate), it matches or beats an unconstrained CNN on a collapse-robust accuracy metric (centre), and its rollout error stays bounded over horizon while the steerable baseline diverges (right, log scale).

(E7) The certificate becomes task competence. Run through a G -equivariant planner (A5), the prediction certificate becomes an *orbit-invariant control* certificate: closed-loop pose error is flat over the orbit to the floating-point floor (ratio 1.000, 3 seeds) while a $4.3\times$ -larger non-equivariant controller degrades out of the training wedge. The flatness is exact because (A5) makes the planner G -equivariant and the planning cost G -invariant, so the closed-loop trajectory inherits the encoder’s orbit-equivariance — Theorem A under the closed-loop clause.

(E8) The certificate is an epistemic drive: certificate-driven active inference. Because the certified region is *computable* (Algorithm 1), an agent can act to **expand** it — a provable alternative to curiosity’s “reduce prediction error”. On a $\mathbb{Z}_2^7$ compositional world whose action space mixes 7 true generators with noisy high-error distractors, an explorer that maximizes certified-region growth certifies all $2^7=128$ compositions in exactly 7 observations (the generator basis — each generator unlocks an exponential swath, Lemma 1), whereas an error-curiosity explorer is lured by the irreducible-noise distractors and certifies only 1% at the same budget (3 seeds; random $\sim 2\%$). So “expand your certified region” is a **noise-immune** epistemic drive that beats prediction-error curiosity — the classic *noisy-TV* failure mode. *Honest scope*: against *raw-error* curiosity (a sophisticated information-gain agent would also avoid aleatoric noise); the certificate supplies that immunity **for free and provably**, no noise model — a toy demonstration of the principle, not a benchmark.

(E2 supplementary figures.)

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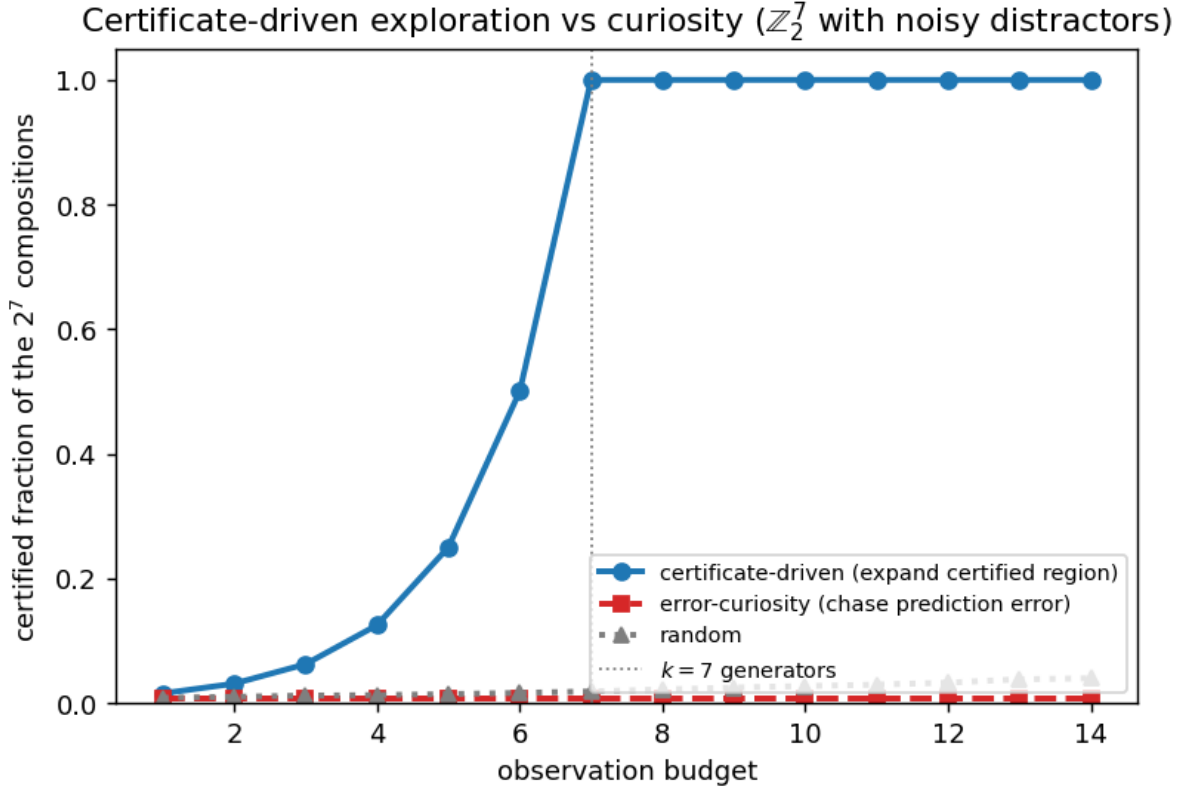


Figure 11: Certificate-driven active inference (E8). On a $\mathbb{Z}_2^7$ compositional world with noisy distractor actions, an explorer that maximizes the *certified region* certifies all 128 compositions in exactly 7 observations (the generator basis), while a prediction-error-curiosity explorer is lured by the high-error distractors and certifies almost nothing at the same budget.

The certified horizon law on a learned model of REAL chaotic dynamics (Lorenz)

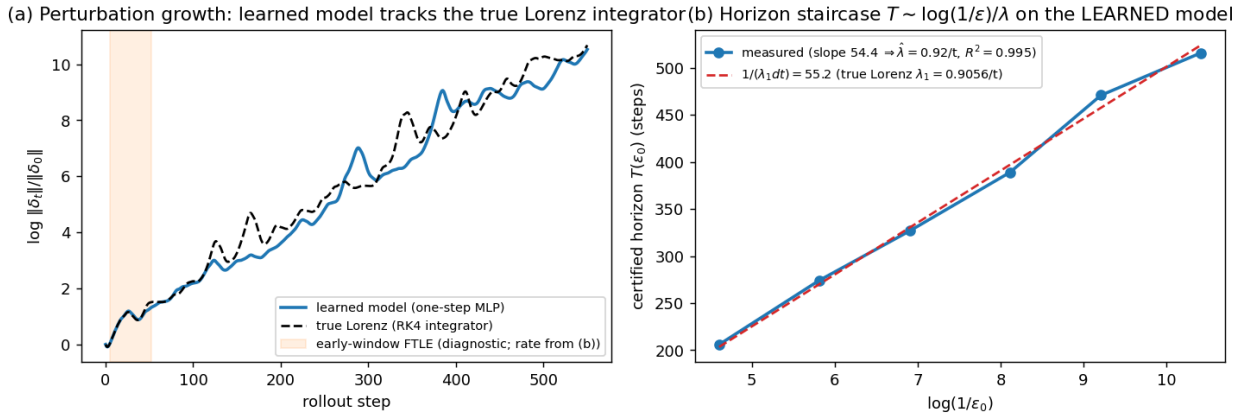


Figure 12: The horizon staircase on a learned model of real chaotic dynamics (E2, Lorenz). **Left:** the learned one-step model’s perturbation growth tracks the true Lorenz integrator over 550 steps. **Right:** the certified horizon $T(\epsilon_0)$ on the *learned* model is linear in $\log(1/\epsilon_0)$ ($R^2=0.995$), and the measured slope sits on the prediction $1/(\lambda_1 dt)$ from the textbook Lorenz exponent — Theorem B’s law lifted to a learned model of a genuinely chaotic system (Proposition 7(a)).

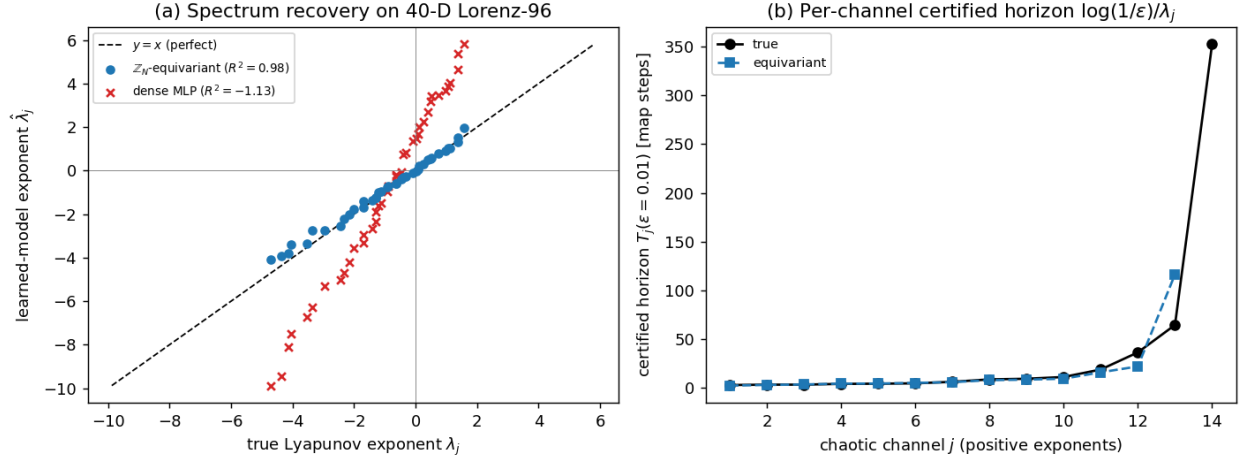


Figure 13: High-dimensional spectral horizon (E2, 40-D Lorenz-96). **Left:** recovered vs. true Lyapunov exponent for all 40 channels — the $\mathbb{Z}_N$ -equivariant cyclic-conv (blue) lies on $y = x$ ($R^2=0.98$); the dense MLP (red $\times$) is scattered far off ($R^2=-1.1$). **Right:** per-channel certified horizon $\log(1/\epsilon)/\lambda_j$: the equivariant model tracks the truth across the spectrum. Structure recovers the high-dimensional spectral horizon a dense model of equal data cannot.

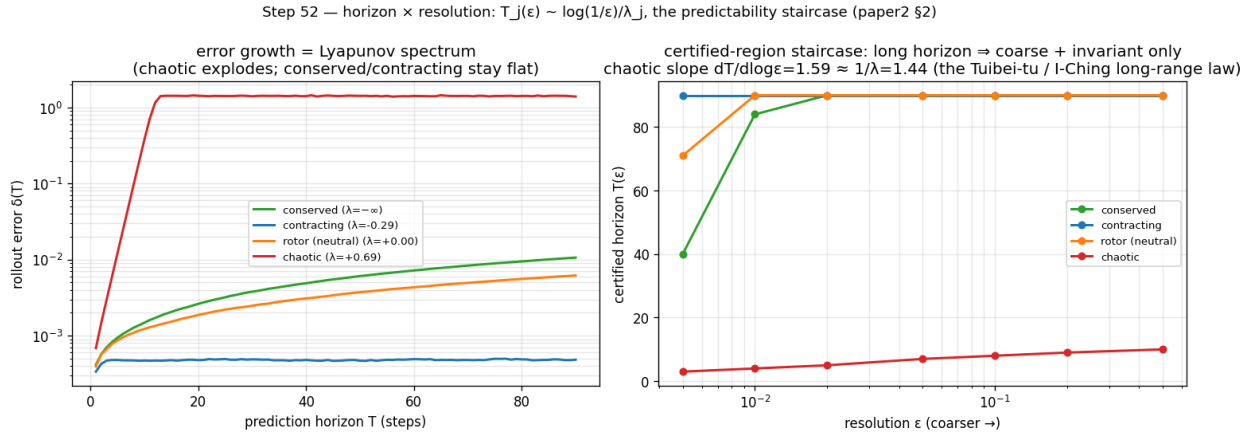


Figure 14: The horizon $\times$ resolution staircase on a controlled spectrum (E2). The measured certified horizon per channel as the demanded resolution ϵ sharpens, recovering $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$ (slope 1.3–1.6): chaotic channels are certified to a few steps, slow/invariant channels to dozens.

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